Katelyn Gan

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EDUCATION

Massachusetts Institute of Technology B.S. Double Major in Mathematics and Computer Science Dec. 2028 (Expected)

Relevant Coursework: Probability and Random Variables, Fundamentals of Statistics, Fundamentals of Programming, Design and Analysis of Algorithms, Introduction to Machine Learning, Linear Algebra **GPA: 5.0/5.0**

WORK & RESEARCH EXPERIENCE

Backend Software Engineering Intern | *Thinkstruct, Boston, MA* **June 2026 – August 2026**

- Built distributed data scraping and ingestion AI pipelines for 5M+ international patent documents, including structured parsing, validation, and infrastructure cost modeling.
- Scaled backend to multi-worker concurrency via Postgres migration and engineered an async LLM document generation system (S3, thread pools), eliminating request timeouts.

MIT Undergraduate Researcher | *PI: Kaiming He, CS and Artificial Intelligence Lab (CSAIL)* **Dec. 2025 – Present**

- Developed 10+ multimodal reasoning benchmarks (30K+ samples/task) and a resumable, diffusion-based data augmentation pipeline on SLURM GPU clusters.
- Tuned open multimodal models using LoRA and deployed LLM-as-judge verifiers to filter generated data at scale.
- Optimized Vision Transformers on the Abstraction and Reasoning Corpus using self-supervised learning, rotary positional embedding, and test-time training on Cloud TPU clusters.

MIT PRIMES-USA Math Researcher | *Mentor: Yunseo Choi, Harvard University* **Jan. 2024 – Feb. 2025**

Simulated lattice-based combinatorial games in Python and proved open conjectures related to game-theoretic behavior.

SELECTED PUBLICATIONS

Xiaoman Ding, Keya Hu, **Katelyn Gan**, Victor Yin, Kaiming He, "Natural Image Pretraining Improves Abstract Reasoning", Accepted to the European Conference on Computer Vision (ECCV), 2026

Yunseo Choi, **Katelyn Gan**, Andrew Li, Tiffany Zhu, "*Set Partitions that Require a Maximum Number of Sorts Through the aba-avoiding Stack,*" Enumerative Combinatorics and Applications, Volume 5:1, Article S2R1, 2025.

Yunseo Choi and **Katelyn Gan**, "*Ungar Games on the Young-Fibonacci and the Order Ideals of Shifted Staircase Lattices,"* submitted for publication, oral presentation at Joint Mathematics Meetings, Seattle, WA, 2025.

Christopher Bao, Giulio Cerbai, Yunseo Choi, **Katelyn Gan**, Owen Zhang, "*On a Conjecture on Pattern-avoiding Machines,*" Annals of Combinatorics, 2024. Also presented at Joint Mathematics Meetings, San Francisco, CA, 2024.

Katelyn Gan and Vinesh Maguire Rajpaul, "*A Computer Vision Approach to Radial Velocity Extraction for Exoplanet Detection,"* 2023 IEEE MIT URTC, Cambridge, MA, USA, 2023.

OTHER INDEPENDENT PROJECTS

arXiv AI Daily — Automated Research Intelligence System (GitHub)
Engineered a scalable pipeline to scrape, embed, and rank arXiv cs.AI papers using FAISS and OpenAI embeddings.

A Toy Language Model — GPT-Style Transformer
Built a PyTorch GPT-style model featuring multi-head self-attention and retrieval-augmented generation via FAISS.

TECHNICAL SKILLS

Languages: Python, C++, SQL, Java, MATLAB, JavaScript

Frameworks: PyTorch, JAX, OpenCV, FAISS, OpenAI API, React, Node.js, Firebase

Infra & Tools: GCP, Distributed Training, TPUs (v5/v6), NVIDIA GPUs, Linux, Git, LaTeX

SELECTED AWARDS & HONORS

MIT Pozen Fellowship ('26) | Citadel Discover ('26) | Jump Trading Early Access ('26) | Jane Street WiSE ('25 & '26) | USAAIO Gold ('25) | USAMO & USAPhO Honorable Mention ('24) | ISEF 4th Place Grand Prize ('24)